

Ramsey-Cass-Koopmans (RCK) Model

1 Motivation

Recall that one of the main assumption of the Solow model was that the savings rate was constant. No household optimization was involved, making welfare analysis impossible. Ramsey model allows household optimization into the growth model. As it turns out, British philosopher and mathematician Frank Ramsey solved this problem in 1928. Later in 1960 in the 1960s David Cass and Tjalling Koopmans independently revived Ramsey's contribution. It has since become ubiquitous and, under the grand moniker of Neoclassical Growth Model (NGM), it is the foremost example of the type of dynamic general equilibrium model upon which the entire edifice of modern macroeconomics is built.

2 The Economy

2.1 Simplifying assumptions

To make the problem manageable, we will assume that there is one representative household, all of whose members are both consumer and producer, living in a closed economy (we will lift this assumption in the next chapter). There is one good and no government. Each consumer in the representative household lives forever, and population growth is $n > 0$ as before. All quantities in small-case letters are per capita. Finally, we will look at the problem as solved by a benevolent central planner who maximises the welfare of that representative household, and evaluates the utility of future consumption at a discounted rate ρ .

2.2 Social Planner's Problem Vs. Market Problem

- We first discuss Social Planner's Problem (SPP) in lieu of market problem because it is easier to solve and it provides the same results as the decentralised equilibrium, as long as both production and consumption sets. This is due to welfare theorems:
 - First Welfare Theorem (FWT): A competitive equilibrium is Pareto Optimal.
 - Second Welfare Theorem (SWT): All Pareto Optimal allocations can be decentralised as a competitive equilibrium under convexity assumptions.
- In the social Planner's Problem, the planner determines the allocation while in the market problem allocations are determined by prices.
- A market problem has three components: (i) Household's Problem, (ii) Firms problem, and (iii) Market clearing conditions (equilibrium). The social planner's problem is similar to household's problem except social

⁰I am truly grateful to my students who helped me prepare this handout. This handout is heavily based on Macroeconomics textbooks written by Campante et al. (2021) and Barro & Sala-i-Martin (2004) and, therefore, does not reflect my original scholarship.

planner faces resource constraint as opposed budget constraint of the representative household in a market economy.

- Note that prices do not enter in a Social Planner's Problem because he does not need market (i.e., prices) to redistribute resources.

3 The Social Planner's Problem (SPP)

The social planner maximizes

$$\int_0^{\infty} u(c_t) e^{nt} e^{-\rho t} dt \quad (1)$$

subject to the resource constraint:

$$Y = F(K, L) = C + I \quad (2)$$

where ρ is the rate of time preference. A $\rho = 0$ implies that the household does not distinguish between current and future consumption while a $\rho > 0$ indicates the household discounts future utility (does not want to wait to consume). We assume that the utility function has standard properties:

- Strictly increasing: $u'(c) > 0$ and $u''(c) < 0$
- Maintains Inada conditions: $\lim_{c \rightarrow \infty} u'(c) = 0$ and $\lim_{c \rightarrow 0} u'(c) = \infty$
- Twice continuously differentiable(C^2): the second derivative exists and is continuous.

The production function $F(K, L)$ is Neoclassical (See my note on the Solow Model for detail).

Recall that the constant returns to scale property allows us to write the aggregate production function in intensive form. Using CRS,

Resource constraint per capita:

$$\begin{aligned} \frac{Y}{L} = y &= \frac{C}{L} + \frac{\dot{K}}{L} \\ \Rightarrow y &= f(k) = c + \frac{\dot{K}}{L} \end{aligned}$$

Since,

$$\begin{aligned} k &= \frac{K}{L} \\ \Rightarrow \log k &= \log K - \log L \\ \Rightarrow \frac{d \log k}{dt} &= \frac{d \log K}{dt} - \frac{d \log L}{dt} \\ \Rightarrow \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - \frac{\dot{L}}{L} \\ \Rightarrow \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - n \\ \Rightarrow \dot{k} &= \frac{\dot{K}}{K} \times \frac{K}{L} - nk \\ \Rightarrow \dot{k} &= \frac{\dot{K}}{L} - nk \end{aligned}$$

$$\begin{aligned}\therefore y = f(k) &= c + \dot{k} - nk \\ \Rightarrow \boxed{\dot{k} = f(k) - c - nk} &\rightarrow \text{Resource constraint in per capita form}\end{aligned}$$

∴ The Social Planner's Problem (SPP):

$$\begin{aligned}\text{Max } & \int_0^\infty u(c_t) e^{-\rho t} e^{nt} dt \\ \text{s.t. } & \dot{k}_t = f(k_t) - c_t - nk_t \\ & k_0 > 0\end{aligned}$$

3.1 Current Value Hamiltonian and Maximum Principle

$$\mathcal{H}_t = u(c_t) e^{nt} + \lambda_t \left(f(k_t) - nk_t - c_t \right)$$

Maximum Principle:

$$\frac{\partial \mathcal{H}_t}{\partial c_t} = e^{nt} u'(c_t) = \lambda_t \quad (1)$$

$$\frac{\partial \mathcal{H}_t}{\partial K_t} = -\dot{\lambda}_t + \rho \lambda_t = \lambda_t \left(f'(K_t) - n \right)$$

$$\Rightarrow -\dot{\lambda}_t = \lambda_t \left(f'(k_t) - n \right) - \rho \lambda_t$$

$$\Rightarrow \frac{\dot{\lambda}_t}{\lambda_t} = - \left(f'(k_t) - n \right) + \rho \quad (2)$$

$$\underline{TVC} : \lim_{t \rightarrow \infty} \lambda_t e^{-\rho t} k_t = 0 \quad (3)$$

3.2 Solving for BGP

From (1)

$$\begin{aligned}e^{nt} u'(c_t) &= \lambda_t \\ \log \lambda_t &= nt + \log u'(c_t) \\ \frac{d \log \lambda_t}{dt} &= n + \frac{1}{u'(c_t)} u''(c_t) \frac{dc}{dt} \\ \Rightarrow \frac{\dot{\lambda}_t}{\lambda_t} &= n + \frac{u''(c_t)}{u'(c_t)} \dot{c}_t \\ \Rightarrow \frac{\dot{\lambda}_t}{\lambda_t} &= n + \left(\frac{u''(c_t)}{u'(c_t)} c_t \right) \frac{\dot{c}_t}{c_t} \\ \Rightarrow \frac{\dot{\lambda}_t}{\lambda_t} &= n - \left(\frac{-u''(c_t)}{u'(c_t)} c_t \right) \frac{\dot{c}_t}{c_t} \quad (4)\end{aligned}$$

$$\text{Define: } \left(\frac{-u''(c_t)}{u'(c_t)} c_t \right)^{-1} = \sigma, \text{ IES.}$$

$$\text{Then, } \frac{\dot{\lambda}_t}{\lambda_t} = n - \frac{1}{\sigma} \frac{\dot{c}_t}{c_t} \quad (5)$$

By using (5) and (2),

$$\begin{aligned} n - \frac{1}{\sigma} \frac{\dot{c}}{c} &= -f'(k_t) + n + \rho \\ \Rightarrow \frac{\dot{c}_t}{c_t} &= \sigma \left(f'(k_t) - \rho \right) \end{aligned} \tag{6}$$

(6) is our Euler Equation (EE) or simply Keynes-Ramsey rule.

Interpretation of the Euler Equation:

When the household saves one unit of output, it goes to investment, which converts into output given by $f'(k)$. But it is worth less by a factor ρ . The household will only postpone consumption i.e., save if $f'(k) > \rho$.

And σ gives the elasticity of intertemporal substitution (EIS) of consumption. This is simply the elasticity of marginal utility of consumption.

3.3 Dynamics of the model

Two ODES and Terminal conditions:

$$\dot{k} = f(k) - c - nk$$

$$\text{and } \frac{\dot{c}}{c} = \sigma(f'(k) - \rho)$$

$$\text{and TVC: } \lim_{t \rightarrow \infty} \lambda_t e^{-\rho t} k_t = 0, k_0 > 0$$

Law of motion of capital (the resource constraint), Euler equation, and the boundary conditions (initial condition and the TVC) will categorize the solution. Once we get k_t and c_t , we can solve for any remaining variables.

Balanced Growth Path (BGP):

In the BGP, $\dot{k} = 0 \Rightarrow$

$$c^* = f(k^*) - n(k^*)$$

Dynamics of k:

For a given k , the level of c that implies $\dot{k} = 0$ is given by $c = f(k) - nk$. And $\dot{k} = 0$ when c equals the difference between output and break-even investment (given by nk line). This value of c is increasing in k until c reaches the maximum level.

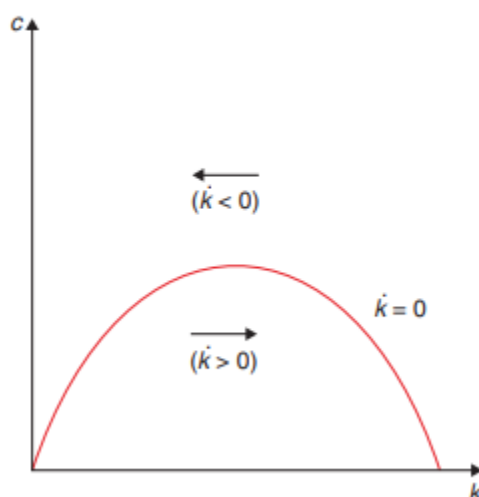
Now,

$$\max_k c = f(k) - nk, \text{ implies}$$

$$\text{FOC, } \frac{dc}{dk} = 0 \Rightarrow f'(k) = n \text{ (golden rule)}$$

So, c is increasing in k until $f'(k) = n$, and then c decreases. Hence, we know a hump- shape $\dot{k} = 0$ locus.

Dynamics of capital



Suppose that c' is the level of consumption that keeps k constant (on the $\dot{k} = 0$ line). Now, if the actual consumption $c > c'$, the consumer will not have adequate saving s to maintain existing level of capital stock. Hence, $\dot{k} < 0$ for any value of c above the $\dot{k} = 0$ locus. Conversely, if c is below $\dot{k} = 0$ line, $\dot{k} > 0$.

Dynamics of c :

From EE,

$$\frac{\dot{c}}{c} = \sigma(f'(k) - \rho)$$

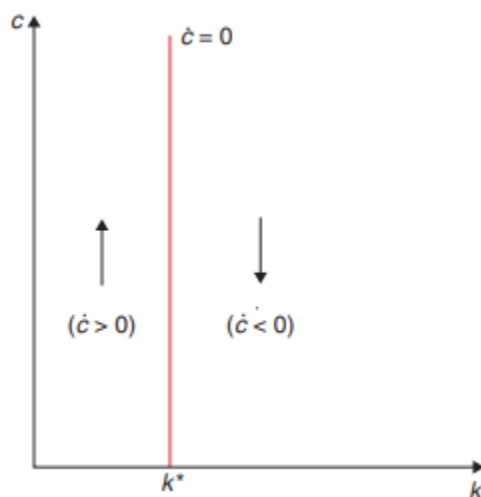
in the BGP,

$$\begin{aligned} \frac{\dot{c}}{c} &= 0 \Rightarrow \\ f'(k) &= \rho \end{aligned}$$

let k^* denote this level of capital.

Note that $\frac{\dot{c}}{c} = 0$ pins down the capital stock. So, $\dot{c} = 0$ is a vertical line in (c, k) space.

Dynamics of consumption



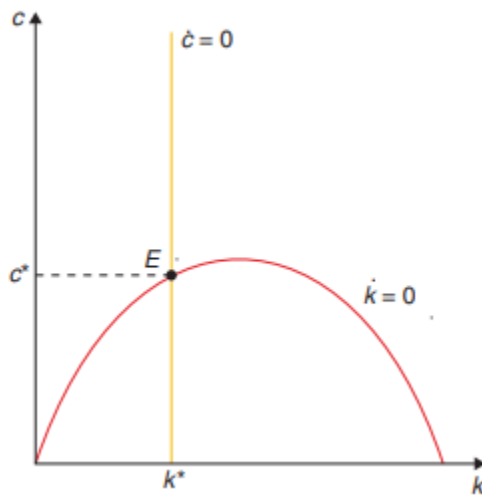
Now, if actual $k > k^*$, then $f'(k) < \rho$, hence, $\dot{c} < 0$ (because of inada condition), and if $k < k^*$, $\dot{c} > 0$.

Steady State:

Since at BGP $\dot{k} = 0$ and $\dot{c} = 0$, the intersection of $\dot{k} = 0$ and $\dot{c} = 0$ locii along with the terminal conditions (TVC and initial value) will fully characterize the steady state solution.

The following figure combines these two locii.

Steady state



Note that k^* (when $\dot{c} = 0$) is less than k_{gold} . Because k^* requires $f'(k) = \rho$ and k_{gold} requires $f'(k_g) = n$. Since $k_{\text{gold}} > k^*$, this requires $f'(k_g)$ at $k_{\text{gold}} < f'(k)$ at k^* . This implies $n < \rho$. (and it holds, by assumption).

Why $n < \rho$?

Recall that the lifetime utility

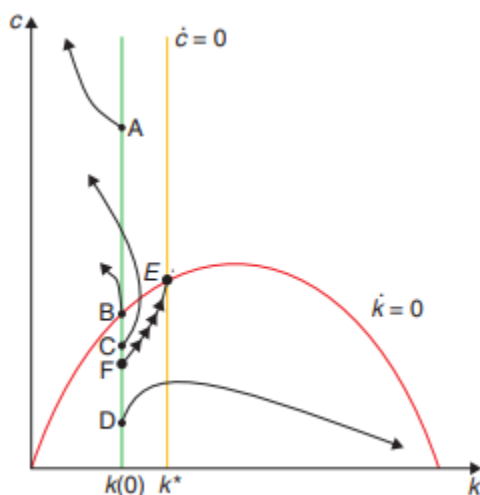
$$v = \int_0^{\infty} u(c) e^{-\rho t} e^{nt} dt$$

if $n > \rho$, then $u(c)$ will explode, and there will be no maximum utility. Hence, for finiteness of utility, we assume $n > \rho$

Initial Value of c:

The Social Planner is given the initial value of $k > 0$. But the consumer will determine the initial value of c .

Now we assume $k(0) < k^*$, for simplicity. We need to trace the trajectories of k and c for various initial values of c . The trajectory that satisfies both ODEs and boundary conditions (TVC) will be the feasible trajectory. Now consider various values of c_0 for a given initial value k_0 .



Now if $c(0) = D$, consumption is too low, both c and k initially positive.

Once k crosses k^* , $\dot{c}$ becomes negative, but $\dot{k}$ remains positive. The economy moves down and to the right.

If $c_0 = C$, both $\dot{c}$ and $\dot{k}$ are initially positive, but $\dot{k}$ becomes negative once c crosses $\dot{k} = 0$ line. The economy eventually moves up and to the left.

Since $\dot{k}$ and $\dot{c}$ are both continuous functions, there exists some critical point say $c(0) = F$ such that the economy converges to the stable point E , where both c and k are constant.

Again if $c(0) = B$ or $c(0) = A$, $\dot{c} > 0$ which reduces capital and economy moves up to the left.

All these various trajectories satisfy the two ODEs. Does this mean all of these are possible?

The answer is **NO**.

If $c(0) > F$, k continuously decreases while c continuously increases. At some point k will eventually be zero. For two ODEs to continue to be satisfied, c must continue to rise and k must continue to fall and become negative, which violates the resource constraint (and also violates TVC). Therefore, any initial value for c above F is not possible.

On the other hand, if $c(0) < F$, then eventually k becomes very large and c becomes zero.

Now TVC says,

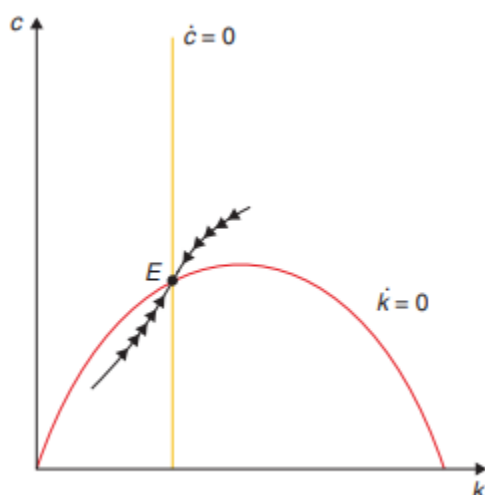
$$\begin{aligned}\lim_{t \rightarrow 0} \lambda_t e^{-\rho t} k_t &= 0 \\ \Rightarrow \lim_{t \rightarrow 0} u'(c) k_t &= 0\end{aligned}$$

Since $u'(c) \neq 0$, TVC does not hold.

Hence, any initial value of $c(0) < F$ is not possible.

Finally, if $c(0) = F$, then c and k continuously rise and converge to k^* . At k^* , $\lim_{t \rightarrow 0} \lambda_{t \rightarrow 0} k_t = 0$ because it is an optimality condition that gives k^* .

Hence, there is a unique initial value of $c(0) = F$ that is consistent with household's intertemporal optimization. The function that gives this initial value of c as a function of k is known as saddle path.



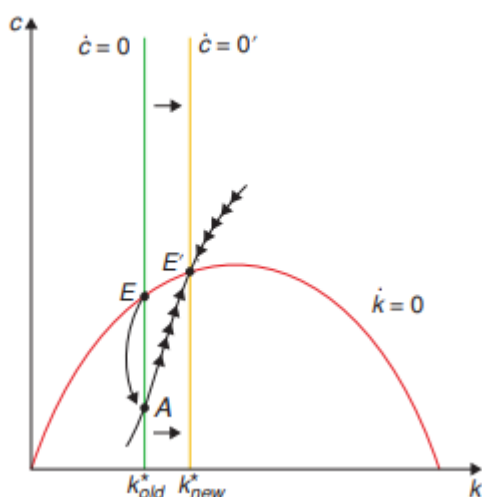
In short, the first two dynamic equations provide the dynamics at any point in the (c, k) space, but only the TVC allows us to choose a single path that we will use to describe our equilibrium dynamics.

4 Effects of Shocks

4.1 A permanent increase in discount rate/fall in ρ

Since $\frac{\dot{c}}{c} = \sigma \left(f'(k) - \rho \right)$, consumption immediately reduces to A (See figure below). Since both $\dot{c}$ and $\dot{k}$ are positive at A , the economy converges to E' . Both c and k eventually rises.

A permanent increase in the discount rate



5 The Equivalence with the decentralized equilibrium

In a decentralized economy We have households and firms (owned by households) who independently make their decisions in a perfectly competitive environment. Recall that a market problem has three components:

- Household's problem
- Firm's problem
- Market clears (Market equilibrium)

5.1 Household's problem:

In a market economy, the representative household maximizes lifetime utility subject to budget constraint (given other restrictions). That is,

$$\int_0^{\infty} u(c_t) e^{nt} e^{-\rho t} dt$$

s.t. $\dot{A} = w_t L_t - c_t L_t + r A_t$

Where A denotes aggregate asset (e.g., bond). Assuming capital and assets are perfect substitutes, the household's saving converts into bond which gives interest at a rate r . Household's saving ($\dot{A}$) is given by the difference between earnings (from labor and non-labor income) and consumption.

Budget Constraint in per capita form:

$$\begin{aligned} \text{Since let } a &\equiv \frac{A}{L} \\ \Rightarrow \log a &= \log A - \log L \\ \Rightarrow \frac{d \log a}{dt} &= \frac{d \log A}{dt} - \frac{d \log L}{dt} \\ \Rightarrow \frac{\dot{a}}{a} &= \frac{\dot{A}}{A} - n \\ \Rightarrow \dot{a} &= \frac{\dot{A}}{A} \frac{A}{L} - na \\ \Rightarrow \dot{a} &= \frac{\dot{A}}{L} - na \\ \Rightarrow \frac{\dot{A}}{L} &= \dot{a} + na \\ \text{Hence, } \dot{A} &= w_t L_t + r A_t - c_t L_t \\ \Rightarrow \frac{\dot{A}}{L} &= w + r \frac{A}{L} - c \\ \Rightarrow \dot{a} + na &= w + ra - c \\ \Rightarrow \boxed{\dot{a} = (r - n)a + w - c} \end{aligned}$$

Restriction on borrowing

Households cannot pursue the easy path of getting arbitrarily rich by borrowing money and borrowing even more to pay for the interest owed on previously contracted debt. If possible that would be the optimal solution. However, the market will not allow these Ponzi schemes, so we impose this as a constraint on household behaviour.

Household's problem:

A complete household problem is given by:

$$\begin{aligned}
& \text{Max} \int_0^\infty u(c_t) e^{-\rho t} e^{nt} dt \\
& \text{s.t. } \dot{a} = (r - n)a + w - c, \\
& \text{NPG: } \lim_{t \rightarrow \infty} a_t e^{-(r-n)t} \geq 0, \\
& a_0 > 0
\end{aligned}$$

State variable: a_t and control variable: c_t

Note on No-Ponzi game condition (NPG):

The no-Ponzi condition is very similar to the TVC in the social planner's problem, but they are different. We did not need no-ponzi game condition for the SPP because there was nobody to borrow from. No-ponzi is a constraint (restriction) on optimization, but TVC is an optimality condition.

5.1.1 Current value Hamiltonian and the maximum principle:

The current value Hamiltonian and the corresponding maxim principles become:

$$\mathcal{H} = u(c_t) e^{nt} + \lambda_t \left((r - n)a + w - c \right)$$

Maximum principle:

$$\frac{\partial \mathcal{H}}{\partial c} = 0 \Rightarrow u'(c_t) e^{nt} = \lambda_t \quad (1)$$

$$\frac{\partial \mathcal{H}}{\partial a_t} = -\dot{\lambda}_t + \rho \lambda_t = (r - n) \lambda_t \quad (2)$$

$$\underline{\underline{\text{TVC}}}: \lim_{t \rightarrow \infty} \lambda_t e^{-\rho t} a_t = 0 \quad (3)$$

From (1),

$$\begin{aligned}
u'(c_t) e^{nt} &= \lambda_t \\
\Rightarrow \log \lambda_t &= \log u'(c) + nt \\
\Rightarrow \frac{d\lambda_t}{dt} &= \frac{1}{u'(c)} u''(c) \frac{dc}{dt} + n \\
\Rightarrow \frac{\dot{\lambda}_t}{\lambda_t} &= \left(\frac{u''(c) \cdot c}{u'(c)} \right) \frac{\dot{c}}{c} + n
\end{aligned} \quad (4)$$

From (2):

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial a_t} &= -\dot{\lambda}_t + \rho \lambda_t = (r - n) \lambda_t \\
\Rightarrow \frac{-\dot{\lambda}_t}{\lambda_t} &= r - n - \rho \\
\Rightarrow \frac{\dot{\lambda}_t}{\lambda_t} &= -r + n + \rho
\end{aligned} \quad (5)$$

Using (4) and (5),

$$\begin{aligned}
 -r + n + \rho &= \left(\frac{u''(c) \cdot c}{u'(c)} \right) \frac{\dot{c}}{c} + n \\
 \Rightarrow r - \rho &= \left(\frac{-u''(c) \cdot c}{u'(c)} \right) \\
 \text{Define : } \left(\frac{-u''(c) \cdot c}{u'(c)} \right) &= \sigma \text{ (IES)} \\
 \Rightarrow r - \rho &= \frac{1}{\sigma} \left(\frac{\dot{c}}{c} \right) \\
 \Rightarrow \boxed{\frac{\dot{c}}{c} = \sigma(r - \rho)} & \tag{6}
 \end{aligned}$$

6 gives the Euler equation for the market problem.

5.2 Firm's problem

Because the representative firm does not own capital, firm's problem is static. Firms maximize profit subject to production technology.

$$\begin{aligned}
 \text{Max } \pi &= F(K, L) - rK - wL \\
 &= L \left(\frac{F(K, L)}{L} - r \frac{K}{L} - w \frac{L}{L} \right) \\
 &= L \left(f(k) - rk - w \right)
 \end{aligned}$$

FOCs :

$$\begin{aligned}
 \frac{\partial \pi}{\partial k} &= L(f'(k) \frac{\partial k}{\partial K} - r \frac{dk}{dK}) = 0 \\
 \Rightarrow \left(f'(k) - r \right) \frac{dk}{dK} &= 0 \\
 \frac{\partial k}{\partial K} \neq 0 &\Rightarrow \\
 f'(k) &= r
 \end{aligned}$$

Again,

$$\begin{aligned}
 \pi &= L \cdot f(k) - rk \cdot L - wL \\
 \frac{\partial \pi}{\partial L} &= f(k) + f'(k) \frac{-K}{L^2} - rk - f'(k) \frac{-K}{L^2} - w = 0 \\
 \Rightarrow f(k) - rk - w &= 0 \\
 \Rightarrow f(k) - f'(k) \cdot k &= w
 \end{aligned}$$

Conclusion # 01: Since $f'(k) = r$, from firm's optimization, Euler equations in both market and social planner's problems are equivalent.

Now, from 1,

$$\begin{aligned}
u'(c_t)e^{nt} &= \lambda_t \\
\Rightarrow \log u'(c_t) + nt &= \log \lambda_t \\
\Rightarrow \frac{d \log u'(c_t)}{dt} &= \frac{\dot{\lambda}_t}{\lambda_t} - n \\
\Rightarrow \frac{\dot{u}'(c_t)}{u'(c_t)} &= -r + n + \rho - n \\
\Rightarrow \frac{\dot{u}'}{u'} &= \rho - r \\
\Rightarrow \dot{u}' &= (\rho - r)u' \\
\Rightarrow \dot{u}' - (\rho - r)u' &= 0
\end{aligned}$$

The above is a homogeneous linear differential equation and, hence, we can separate variables.

Solution:

$$\begin{aligned}
\dot{u}' &= (\rho - r)u' \\
\Rightarrow \frac{du'}{dt} &= (\rho - r)u' \\
\Rightarrow \int \frac{1}{u'} du' &= \int (\rho - r) dt \\
\Rightarrow \log u' &= (\rho - r)t + \log C \\
\Rightarrow u' &= Ce^{(\rho - r)t} \\
\Rightarrow \text{Assuming } C = 1, \quad u' &= e^{(\rho - r)t}
\end{aligned}$$

Conclusion # 02: TVCs in both market and social planner's problems are equivalent.

$$\begin{aligned}
\text{From FOCs, } u'(c)e^{nt} &= \lambda_t \\
\Rightarrow e^{(\rho - r)t}e^{nt} &= \lambda_t \\
\Rightarrow e^{\rho t}e^{-rt}e^{nt} &= \lambda_t \\
\Rightarrow \boxed{e^{(n - r)t} = \lambda_t e^{-\rho t}}
\end{aligned}$$

$\therefore$ TVC:

$$\begin{aligned}
\lim_{t \rightarrow 0} a_t e^{-(r - n)t} = 0 \\
\Rightarrow \lim_{t \rightarrow \infty} a_t \lambda_t e^{-\rho t} = 0
\end{aligned}
\left. \vphantom{\begin{aligned} \lim_{t \rightarrow 0} a_t e^{-(r - n)t} = 0 \\ \Rightarrow \lim_{t \rightarrow \infty} a_t \lambda_t e^{-\rho t} = 0 \end{aligned}} \right\} \text{same TVC}$$

5.3 Market Clearing

We can show market clearing conditions are equivalent to resource constraint in the social planner's problem. (Math Later) In the current setup we have three markets: Goods market, asset market and the market for

capital services. By Walrus law, we need to clear just two markets out of the three. The following are the market clearing conditions in goods and bond markets.

- Goods market:

$$\begin{aligned}\sum (c_i + i_i) &= \sum y_i \\ \Rightarrow C + I &= Y\end{aligned}$$

- Bond market:

$$\sum a_i = 0$$

Note on market clearing:

Since all households are alike, if anyone wants to buy bonds, everybody else does the same. This gives rise to the problem of lack of double coincidence of wants. Therefore, no trade of bonds takes place in equilibrium.

Conclusion# 03: Resource constraint in the social planner's problem is the same as market clearing condition in the market problem.

In conclusion, two problems deliver the same EE, they have the same FOC, the same TVC, and, finally, resource constraint is the same as market clearing condition. Therefore, we can conclude that two problems are equivalent i.e., SPP $\Leftrightarrow$ Market Problem.

6 Concluding Remarks (from Campante et al., 2021)

It can be shown that without technological growth, per capita variables grow at zero rate (same as the Solow model). That is, we are still left with a growth model without long-run growth: it was not the exogeneity of the savings rate that generated the unsatisfactory features of the Solow model when it comes to explaining long-run growth. We will have to keep looking by moving away from diminishing returns or by modelling technological progress. On the other hand, our exploration of the Ramsey model has left us with a microfounded framework that is the foundation of a lot of modern macroeconomics. This is true not only of our further explorations that will lead us into endogenous growth, but eventually also when we move to the realm of short term fluctuations. At any rate, the NGM is a dynamic general equilibrium framework that we will use over and over again.

Even in this basic application some key results have emerged. First, we have the Euler equation that encapsulates how consumers make optimal choices between today and tomorrow. If the marginal benefit of reducing consumption – namely, the rate of return on the extra capital you accumulate – is greater than the consumer's impatience – the discount rate – then it makes sense to postpone consumption. This crucial piece of intuition

will appear again and again as we go along in the book, and is perhaps the key result in modern macroeconomics. Second, in this context there is no dynamic inefficiency, as forward-looking consumers would never choose to oversave in an inefficient way. Most importantly, now we are in possession of a powerful toolkit for dynamic analysis, and we will make sure to put it to use from now on.

.....The End.....